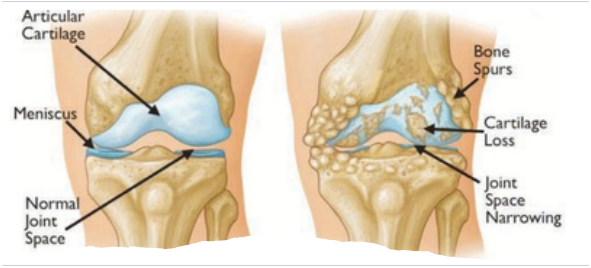
# GENICULAR ARTERY EMBOLIZATION (GAE) for OA of the Knee

- OA of the knee is one of the leading causes of disability and chronic pain worldwide
- Functional limitations impact daily life causing pain, stiffness and difficulty with activities such as walking, climbing stairs and sleeping
- Mental Health and social life are negatively impacted by limitations of OA
- 43% of people with OA are 65 or older and 88% of people with OA are 45 or older
- Prevalence of OA increases with age, and it's more common in women and individuals with higher BMI




## How is GAE performed?

- Femoral or Pedal access
- Select out the Genicular artery
- Embolize

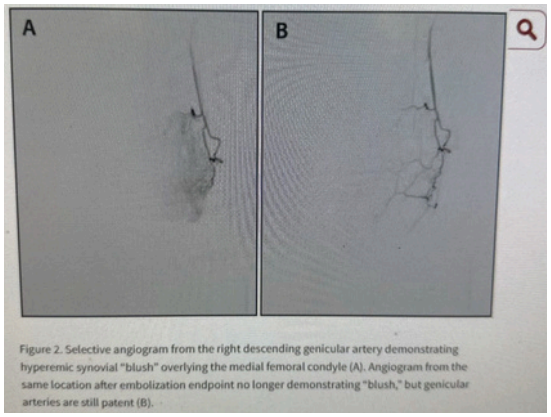
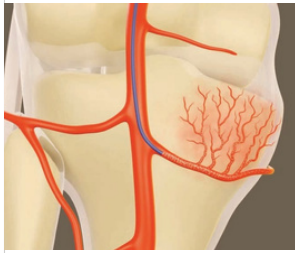
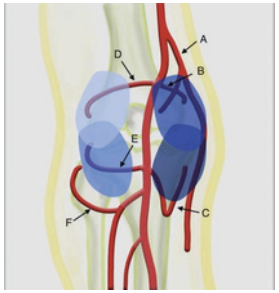


Figure 2. Selective angiogram from the right descending genicular artery demonstrating hyperemic synovial "blush" overlying the medial femoral condyle (A). Angiogram from the same location after embolization endpoint no longer demonstrating "blush," but genicular arteries are still patent (B).





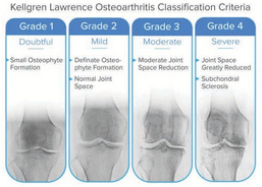

## Pathophysiology of OA

- Wear / tear and Underlying Synovial Inflammation in OA
- Inflation stimulates the release of certain cytokines
- IL1 & IL6 and VEGF
- Leads to Angiogenesis In the Synovium
- Cytokines also cause Synovitis, degeneration of cartilage and osteophyte formation
- Increase in pain fibers leading to pain

## Screening for OA

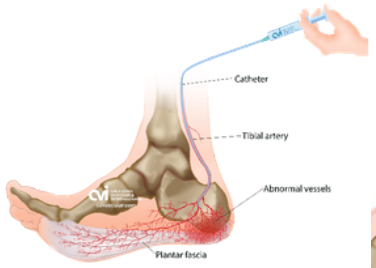
- Refractory to Medications
- VAS Scale Score
- WOMAC Score
- KL Grade
- X-ray of the Knee

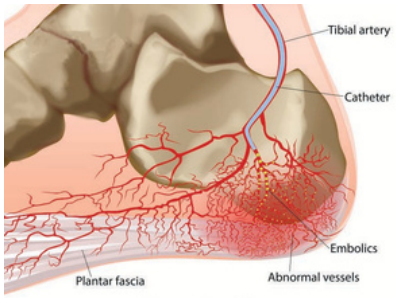


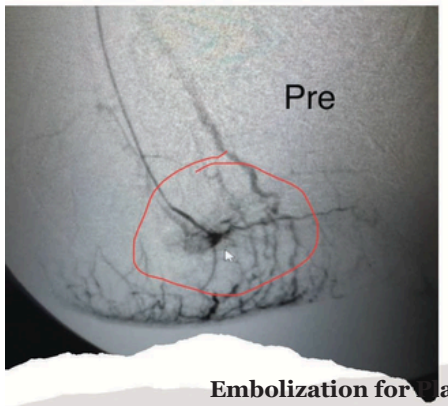

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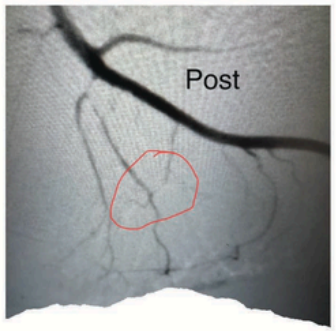
# Embolization for Plantar Fasciitis




Plantar Fasciitis




**Embolization for Plantar Fasciitis**

- Minimally invasive treatment for chronic heel and arch pain from plantar fasciitis.
- Blocks small blood vessels to the inflamed plantar fascia, reducing blood flow, inflammation, and pain.
- Considered when physical therapy, orthotics, and other conservative treatments fail.

### How It Works

- Catheter inserted via ankle or groin using X-ray or ultrasound guidance.
- Tiny embolic particles injected to block targeted blood vessels.
- Reduced blood flow promotes healing and pain relief.

### Benefits

- Minimally invasive: Small puncture, no large incisions or stitches.
- Outpatient: Usually go home same day.
- Quick recovery: Return to activities in days.
- Long-term relief: Targets inflammation source.
- Surgery alternative: Less invasive option.

### Candidates

- Chronic plantar fasciitis unresponsive to conservative treatments.
- Prefer less invasive approach than surgery.

### Candidates

- Consultation: Essential to confirm suitability.
- Risks: Potential complications, as with any procedure.

**We take care of patients with Leg Pain, Leg Swelling, Leg or Foot Ulcers, Neuropathy, and Osteoarthritis of the Knee and Plantar Faciitis.**

Vinod Kumar, MD • Cell (661) 204-2222